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Transport and thermal properties of $(\text{PEO})_n\text{--LiPF}_6$ electrolytes for super-ambient applications

A. Magistris, P. Mustarelli, E. Quartarone*, C. Tomasi

CSTE-CNR and Dipartimento di Chimica Fisica, Università di Pavia, Via Taramelli, 1627100 Pavia, Italy

Abstract

Poly(ethylene oxide) (PEO)-based solvent-free electrolytes may be used for applications above room temperature, e.g. in batteries for electric vehicles. The system PEO--LiPF_6 has to date received only scarce attention because of the poor thermal stability and hygroscopicity of the lithium salt. However, for applications above the melting of PEO crystalline phase, it is possible to consider films with relatively low salt content, which can overcome the above-mentioned problem. In this paper, we perform a thermal and transport characterisation of the system PEO--LiPF_6 . We show that conductivity values above $10^{-4} \Omega^{-1} \text{cm}^{-1}$ can be reached at 70°C for $n=\text{EO}/\text{Li}=30$. The addition of nanoscale silica is shown to influence the thermal decomposition of the salt. © 2000 Elsevier Science B.V. All rights reserved.

Keywords: PEO; LiPF_6 ; Electrolyte; Transport; Thermal properties

1. Introduction

Lithium polymer batteries seem to be very promising for applications not only in electronic portable devices, but also in electric vehicles, because they combine the high energy density of lithium with the processing, cost and geometry flexibility of the polymers [1].

Between the two classes of polymer electrolytes most widely studied in these last years, gel-types and PEO-base solvent free systems, the latter seem to offer better performances above room temperature. In fact around $70\text{--}80^\circ\text{C}$ their conductivity can reach 10^{-3}S cm^{-1} , while their solvent-free nature prevents evaporation phenomena and offers an improved lithium interfacial stability.

However, PEG-based electrolytes still need to be suitably optimised, because of their poor mechanical performances, particularly in the presence of a substantial amount of polymer amorphous phases. In order to eliminate any residual liquid contamination deriving from the casting procedure, the hot pressing technique has been also proposed as an alternative method to prepare films with improved electrochemical properties [2,3]. Moreover, in order to have free-standing films for super-ambient applications, a possibility could be the choice of not plasticising salts and polymer/salt molar ratios far from the eutectic composition.

Several lithium salts with well-defined electrochemical properties are now available. Among them, LiPF_6 is one of the most used as the component in non-aqueous liquid electrolytes for lithium batteries, due to its low cost and good environmental impact. While the literature well defines the behaviour of

*Corresponding author. Fax: +39-0382-507-575.

E-mail address: eliana@chifis.unipv.it (E. Quartarone).

LiPF₆-based organic solutions [4], only few data are available about their use as a solute in poly(ethylene-oxide) and consequently about the thermal and electrochemical properties of the resulting PEO-based electrolytes [5]. There are non-negligible factors which limit the use of this salt in solvent-free systems. Firstly, above 100°C LiPF₆ undergoes thermal decomposition and reduces to LiF [6] with evolution of gaseous by-products. Moreover, the conductivity of these systems is too low for room temperature performances, likely because of poor salt dissociation, contrary to salts with oxygen-containing anions like LiClO₄ or LiN(CF₃SO₂)₂. In 1985 Rietman et al. reported a σ value of 2.5×10^{-7} S/m at 57°C for the complex (PEO)_{4.5}-LiPF₆ [5]. However, it could be interesting to investigate the PEO-LiPF₆ system as a new electrolyte for super-ambient applications, by choosing high molar ratios $n = \text{EO}/\text{Li}$ ($n > 30$) and by working at temperatures above the melting point of PEO, where conductivities higher than 10^{-4} S/cm may be reached. In this way it should be possible to save good mechanical performances without falling into the thermal instability of the salt.

In this paper we present a study on the solvent-free system PEO_{*n*}-LiPF₆, ranging *n* from 100 to 12, in order to individuate suitable compositions for super-ambient applications. Samples with lower *n* displayed segregation phenomena and/or excessive sensitivity to moisture. DSC, TGA and impedance spectroscopy experiments are performed to outline the dependence of thermal and electrical properties on the salt composition. Finally, our attention is focused on their optimisation by the introduction of nanoscale silica. In particular, we investigate the influence of the filler on the stability range of these new electrolytes.

2. Experimental details

2.1. (PEO)_{*n*}-LiPF₆

Films of (PEO)_{*n*}-LiPF₆ were prepared by the conventional solvent casting procedure in a dry box (~10 ppm H₂O), for $n = \text{EO}/\text{Li}$ ranging from 12 to 100. A 0.5 g quantity of dried high molecular weight PEO (MW~ 4×10^6) was dissolved in anhydrous

acetonitrile (Aldrich, H₂O < 5 ppm). Defined amounts of anhydrous LiPF₆ (Aldrich) were then added to the PEO solution, that was finally cast onto a Teflon disk. The resulting films of ~100 μm thickness were characterised both as prepared and after a 2-months-long ageing at room temperature in dry box.

2.2. (PEO)₃₀-LiPF₆ + *X* wt% SiO₂

Fumed SiO₂ (Aldrich, 7 nm) was used in order to obtain films of PEO-based composites. Defined amounts of *filler* (0 < *X* < 20) were added to the PEO-salt solution, prepared as above described, and homogenised with a high speed blender. The resulting solution was then cast onto a Teflon disk to slowly evaporate the solvent.

2.3. Apparatus

DSC measurements have been carried out under N₂ purge on samples of about 20 mg, at 5°C/min, using a DSC 2910 (TA Instruments, USA), controlled by a 2000 Thermal Analyst (TA Instruments).

Thermogravimetric experiments have been carried out at 5°C/min, under N₂ flow, using a 951 TGA (Du Pont, USA).

The conductivity has been measured by impedance spectroscopy (IS), with a Solartron 1255 Frequency Response Analyzer (FRA), in the temperature range 25–130°C and varying the frequency from 1 Hz to 1 MHz. Two heating-cooling cycles were performed to check the reproducibility of the electric behaviour of the electrolytes.

³¹P NMR at the Magic Angle (MAS) spectra were acquired on a AMX400WB spectrometer (Bruker) at a frequency of 161.9 MHz. The spinning rate was 2–3 kHz. The 90° pulse was 5 μs. The spectra are referenced to 85% H₃PO₄ solution.

3. Results and discussion

3.1. PEO_{*n*}-LiPF₆ system

Fig. 1 reports the DSC thermograms of some selected (PEO)_{*n*}-LiPF₆ samples aged for 2 months at room temperature. Beside the sub-ambient glass

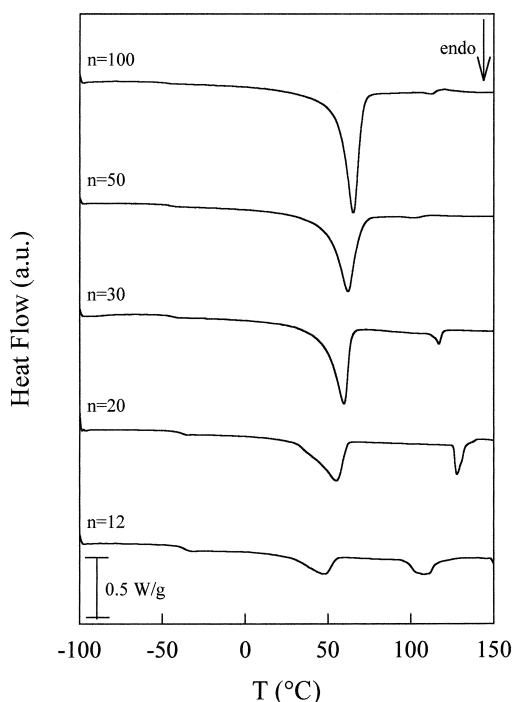


Fig. 1. DSC thermograms of some selected $\text{PEO}_n\text{-LiPF}_6$ films.

transition, T_g , the main thermal feature is an endotherm at $\sim 50^\circ\text{C}$. A second endotherm above 100°C is observed only for $n < 30$. The first endotherm represents the well-known melting of the crystalline phase of PEO, which decreases by increasing the salt content. The second peak is likely correlated to the thermal instability of LiPF_6 . Thermogravimetric experiments upon the salt and the doped matrices have confirmed the nature of such a phenomenon. Fig. 2 shows the TGA curve for the sample $(\text{PEO})_{30}\text{-LiPF}_6$, in which a sharp weight loss is detected just around 100°C , in correspondence with the high-temperature DSC endotherm.

Fig. 3 reports the T_g 's and the PEO melting enthalpy values, ΔH_m , of the $\text{PEO}_n\text{-LiPF}_6$ system in the range $100 \leq n \leq 12$, versus the salt content. The ΔH_m values are normalised to the PEO fraction. A correlation between these two properties is easily observed. As expected for PEO-based electrolytes [7], the glass transition temperature linearly increases with the salt content up to -40°C , reaching the highest value for $n = 12$. This behaviour is justified

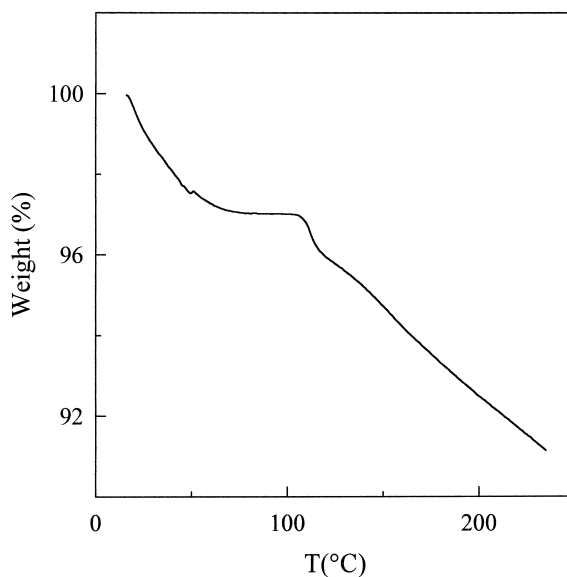


Fig. 2. TGA curve of a sample of $\text{PEO}_{30}\text{-LiPF}_6$.

by the transient cross-linkings among the Li^+ ions and the PEO chains that cause the stiffening of the polymeric network. A similar trend has been recently observed in $\text{PEO}_n\text{-LiClO}_4$ electrolytes and in their composites with $\alpha\text{-Al}_2\text{O}_3$ [8]. The changes in ΔH_m are in reasonable agreement with previous data reported for the homologous system PEO-LiAsF_6 [9].

Fig. 4 shows the behaviour of the conductivity versus salt content at four different temperatures: 20°C , 60°C , 70°C and 80°C . The measurements show that the room temperature conductivity of the system $(\text{PEO})_n\text{-LiPF}_6$ never exceeds $\sim 10^{-8} \text{ S/cm}$, but it abruptly increases up to more than 10^{-4} S/cm above the PEO melting-point at $\sim 60^\circ\text{C}$. A quite surprising effect is that above the PEO melting point there are no remarkable variations of the σ values when decreasing the polymer fraction below $n = 100$, where the conductivity is already two orders-of-magnitude higher than that of pure PEO. This result indirectly suggests that the salt dissociation constant does not change significantly for $n < 100$.

Following the considerations reported in the Introduction, we will focus our attention on the sample with $n = 30$, which represents a good compromise between high conductivity and low salt content.

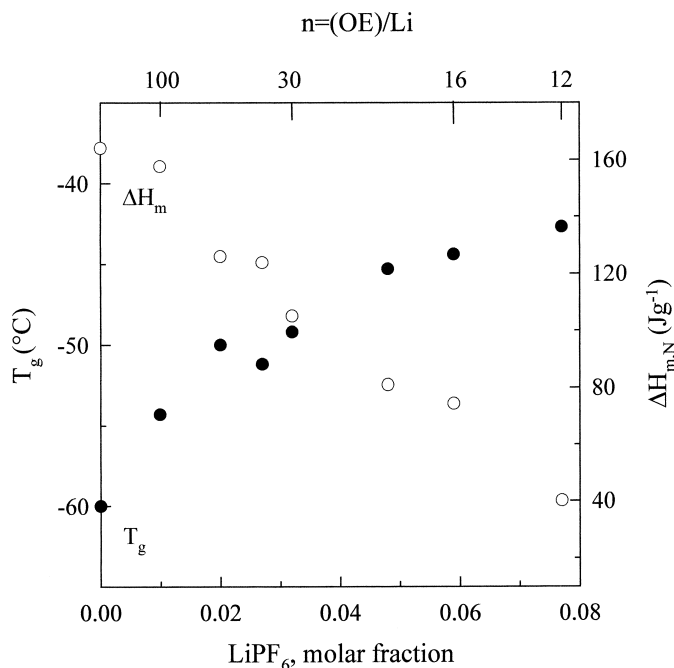


Fig. 3. ΔH_m (○) and T_g (●) values against the salt content of the system $\text{PEO}_n\text{--LiPF}_6$ for $100 \leq n \leq 12$. The values of ΔH_m are normalised to the PEO fraction.

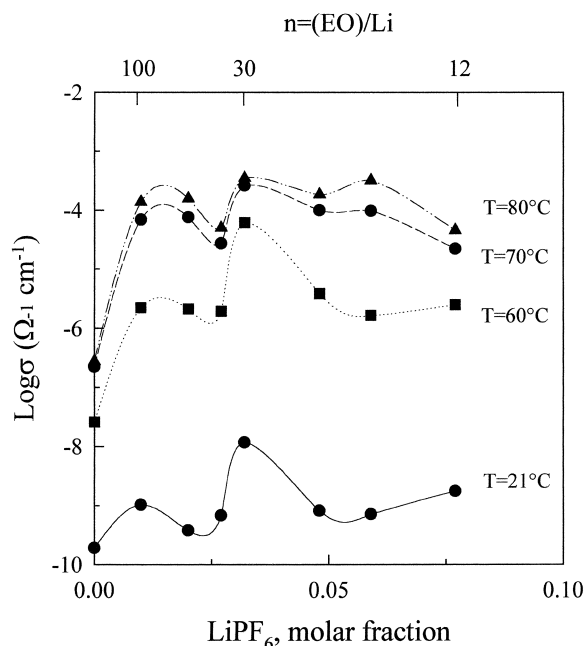


Fig. 4. Conductivity behaviour at different temperatures vs. the salt content for $\text{PEO}_n\text{--LiPF}_6$ electrolytes.

3.2. $\text{PEO}_{30}\text{--LiPF}_6 + X \text{ wt\% SiO}_2$

Fig. 5 shows the DSC thermograms of $(\text{PEO})_{30}\text{--LiPF}_6$ pure (a) and with 5 wt% and 10 wt% of nanoscale silica (b, c). The presence of the filler does not seem to affect substantially the thermal properties of the polymer matrix: in fact, the glass transition temperature of the polymer amorphous phase remains nearly constant by increasing the silica amount, whereas the PEO crystalline phase increases by 5–10%. This last behaviour is in contrast with that generally observed in PEO-based composite polymer electrolytes [10].

On the other hand, silica seems to induce an exothermic process near 120°C, whose intensity is higher in the sample containing 5 wt% of filler, instead of the endotherm of salt decomposition. The nature of such an exothermic peak has been investigated by DSC subsequent heating/cooling/heating cycles up to 200°C, which have confirmed its nature to be a crystallisation process driven by the decomposition of the salt.

This complex behaviour has been investigated by

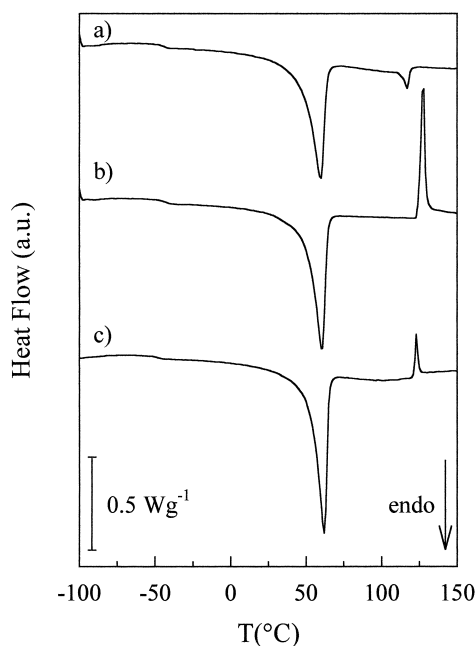


Fig. 5. DSC traces of some films: $\text{PEO}_{30}\text{-LiPF}_6$ (a), $\text{PEO}_{30}\text{-LiPF}_6$ containing 5 wt% SiO_2 (b) and $\text{PEO}_{30}\text{-LiPF}_6$ containing 10 wt% SiO_2 (c).

^{31}P NMR–MAS experiments. Fig. 6 reports the NMR spectra for the sample $(\text{PEO})_{30}\text{-LiPF}_6$ as prepared (a), and for the samples $(\text{PEO})_{30}\text{-LiPF}_6$ (b) and $(\text{PEO})_{30}\text{-LiPF}_6\text{-10 wt% SiO}_2$ (c) after heating up to 130°C . The spectrum of the sample $(\text{PEO})_{30}\text{-LiPF}_6\text{-10 wt% SiO}_2$ as prepared is identical to that of the film without filler. In the spectrum (a) only the PF_6^- structure is observable at ~ 146 ppm (a spin–spin coupling typical of a system AX_6), whereas both the heated samples are characterised by different chemically shifted signals, which correspond to the products derived by the decomposition of the salt. In particular, the NMR spectrum of the treated $\text{PEO}_{30}\text{-LiPF}_6$ film (b) shows a triplet (marked with stars) and a doublet (marked by circles) given by P–F spin–spin couplings in the degradation intermediate structures, $(\text{OPOF}_2)^-$ and $(\text{O}_2\text{POF})^{2-}$, and a less intense singlet at 0 ppm, which is typical of PO_4^{3-} groups. In contrast, in the nanocomposite film (c) the PO_4^{3-} singlet at 0 ppm is by far the main feature, followed by the doublet of $(\text{O}_2\text{POF})^{2-}$. The peak at ~ 10 ppm is a spinning sideband. The decomposition of LiPF_6 is assisted by the presence

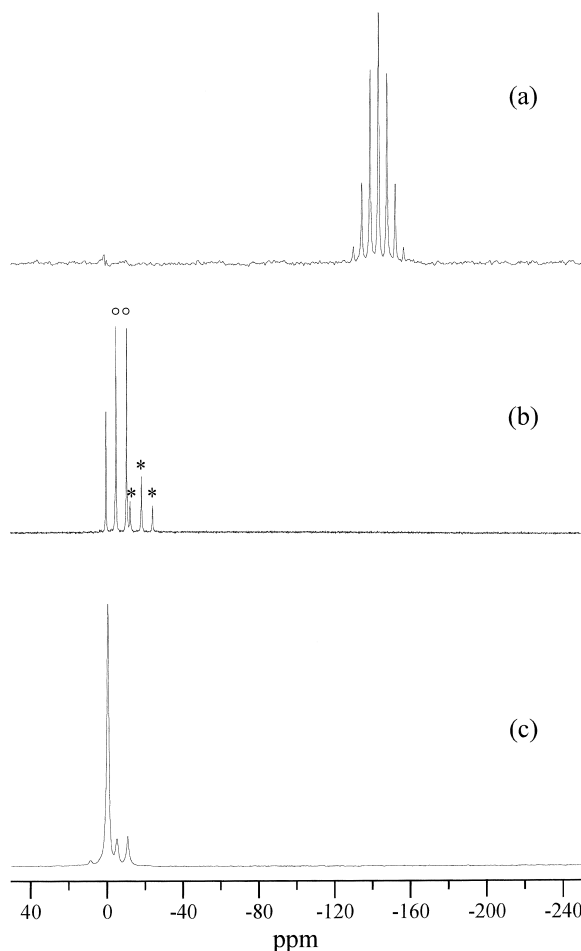
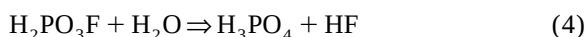
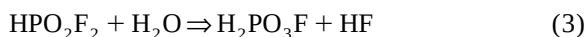
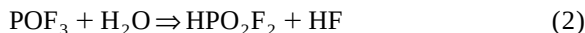


Fig. 6. ^{31}P NMR–MAS spectra of $\text{PEO}_{30}\text{-LiPF}_6$ as prepared (a); $\text{PEO}_{30}\text{-LiPF}_6$ at 130°C (b); $\text{PEO}_{33}\text{-LiPF}_6\text{-10 wt% SiO}_2$ at 130°C (c).

of moisture [5], probably adsorbed on the salt and on SiO_2 , and can be outlined by two steps as reported in the following scheme:



The differences in the degradation final products between pure and nanocomposite systems are likely

due to the presence of the silica and, therefore, to the overall amount of water contaminating the films. In the pure matrix, the decomposition of LiPF_6 is dominated by the first step and no crystallisation is observed. After the introduction of the filler, however, a higher moisture content is available for the salt decomposition, so the intermediate structures completely react to form PO_4^{3-} , which likely crystallises as Li_3PO_4 . However, the most interesting point is that filler addition delays the bulk decomposition of the salt by about 30°C , as demonstrated by TGA measurements on samples with a silica content varying from 0 to 20 wt%. Fig. 7 shows as an example the TGA curves for the matrices containing 0 (a) and 15 wt% (b) of filler.

Fig. 8 shows the Arrhenius plots of the conductivity (first heating scan) of the samples $\text{PEO}_{30}\text{-LiPF}_6$, $\text{PEO}_{30}\text{-LiPF}_6\text{-5 wt\% SiO}_2$ and $\text{PEO}_{30}\text{-LiPF}_6\text{-10 wt\% SiO}_2$. Contrary to systems like $\text{PEO}_8\text{-LiClO}_4$ or $\text{PEO}_8\text{-LiN}(\text{CF}_3\text{SO}_3)_2$ [11,12], the introduction of silica in the $\text{PEO}_{30}\text{-LiPF}_6$ electrolyte depresses the conductivity. This point will be thoroughly discussed in a companion paper [13]. The clear slope change at 120°C in the curve of $\text{PEO}_{30}\text{-LiPF}_6$ (●) is related to salt decomposition, while the well-evident transition at 125°C in the sample con-

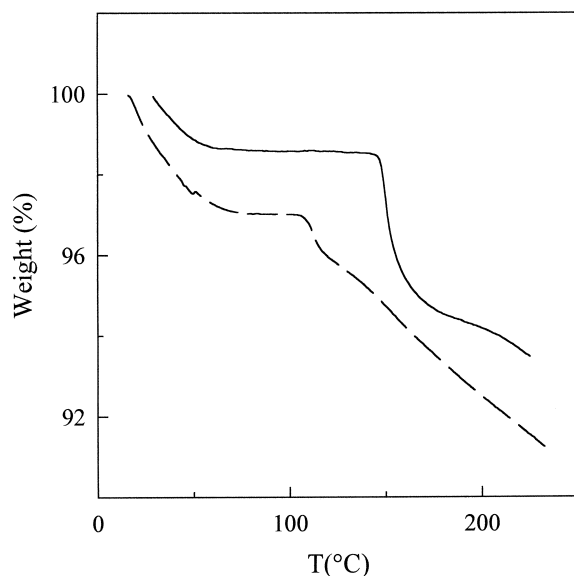


Fig. 7. TGA thermograms of the samples $\text{PEO}_{30}\text{-LiPF}_6$ (a) and $\text{PEO}_{30}\text{-LiPF}_6$ containing 15 wt% SiO_2 (b).

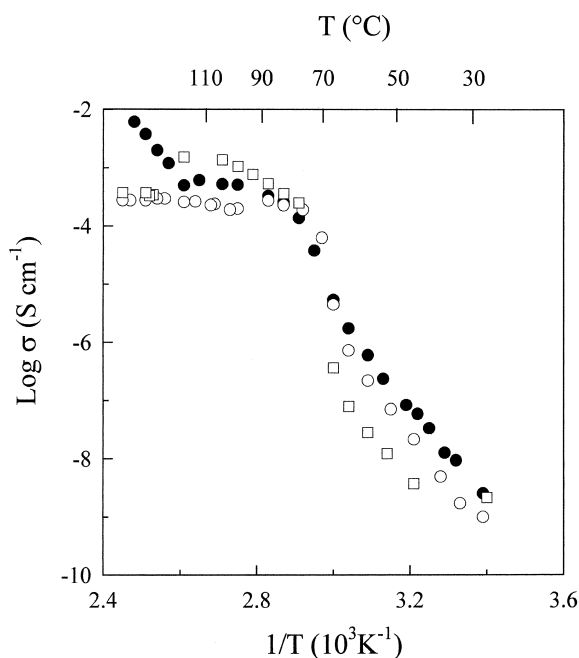


Fig. 8. Arrhenius plots of the conductivity for some selected matrices. (●) $\text{PEO}_{30}\text{-LiPF}_6$; (□) $\text{PEO}_{30}\text{-LiPF}_6\text{-5 wt\% SiO}_2$; (○) $\text{PEO}_{30}\text{-LiPF}_6\text{-10 wt\% SiO}_2$.

taining 5 wt% SiO_2 (□) is related to the exothermic peak observed in the DSC thermogram (see Fig. 4b). This transition involves a slight decrease in conductivity, which is expected following the formation of crystalline Li_3PO_4 . Such an effect is evident only in the sample containing 5 wt% SiO_2 , where the exotherm is more intense.

4. Conclusions

In this paper we have reported a thermal and transport characterization of the system $\text{PEO}_n\text{-LiPF}_6$. The films prepared by solvent casting are thermally stable at least up to $\sim 90^\circ\text{C}$. Conductivity values higher than $10^{-4} \Omega^{-1} \text{cm}^{-1}$ can be reached at 70°C also in the case of relatively low salt contents. The addition of nanoscale silica at the composition $\text{PEO}_{30}\text{-LiPF}_6$ slightly depresses the conductivity. The salt decomposition has been followed by solid-state ^{31}P NMR-MAS. A tentative mechanism is proposed on the basis of preliminary spectroscopic evidences.

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